ENGLISH

Chapter 6

Focus

Our ability to get and stay motivated, to focus, and to learn new things is at the heart of what it is to be successful in life. Not just in school and work, but in creative endeavors and athletics. Consider the effort it takes to study for a big test or remember the material needed for a big presentation, or even Stick with a new approach to a painting or new musical instrument practice, or try to respond to a social situation in a new way. The question then becomes, "How?" How can we get motivated when our energy or natural impulses pull us toward apathy, overthinking, or procrastination? How can we tap into heightened focus, on demand?

How can we learn faster and more effectively? It turns out these are states of mind and body that we can learn to deliberately access. My second year of college is when I first started to discover this. I learned that motivation can happen automatically as the consequence of, say, being excited or intrigued by something but it can also be deliberately fostered. As I mentioned a bit in the introduction, I went from being a good student in my early life to falling behind my peers in high school.

I touched on Why that happened, but I make no excuses. Those were ultimately my own choices and actions. And those challenges both in and out of school, and watching my peers face those challenges too—seeing some soar and thrive while others struggled—would all become a blessing in many ways. It forced me to realize that no one was coming to save me and that I had to change things on my own. That realization scared me, but it's what motivated me to turn my life around.

I never quite had the exact words for what went through my head at that moment until recently, when a very wise friend and mentor said something and I thought, "THAT! That's what I was hearing in my head! THAT!" I can't say I came up with it, but it's something that I have heard in my head in fragments, over and over since I was nineteen. I Still remind myself of it every day: Losers take the negative things that happen to them and the feelings that come from things that suck... and they wallow and they use itfor self- or outward destruction. Winners however take whatever they feel—even if it sucks—and they transmute those feelings into actions that are good for themselves and for the world.

I chose to do something. And through that choice, I learned that while motivation can result in action, it's possible for action to result in motivation too. As a sophomore in college, I decided to study as if my life depended on it. I didn't care if I liked the subject. I decided it made no difference, that eventually I would find the right path if I kept moving forward but I was going to move forward no matter what.

Sure enough, my motivation to learn grew more and more, and morphed into a desire to get a PhD in a topic I fell in love with, to run a lab, to teach, and to build a career in science. Somewhere early along the way that fear I described dissipated. My love of certain topics and You are asking your brain to change, and your brain doesn't want to change. Changing is very energetically demanding for brains. This uncomfortable feeling is something I often refer to as "limbic friction." It's not an oficial scientific term, but it refers to a state of elevated autonomic arousal where you're funneling into focus, and it takes constant or near-constant effort to get there and stay there.

It's a mental edge of sorts. Once I understood that this limbic friction was not only normal, but essential, focus and learning became easier. I started seeking it out. And it felt good. I share this backstory simply to help illustrate that motivation, focus, and learning are states of mind and body.

They have neural and hormonal underpinnings, much in the same way that REM sleep has its neural and hormonal underpinnings, as does stress, and so on. And that's good news, because it makes what might otherwise be nebulous, clear. We know what to aim for. This is a physiology-first approach to focus, motivation, and learning. It's rooted in how we work, which is Why the approach works.

In this chapter you'll learn how to enhance your ability to focus by leveraging that. You'll also learn a powerful tool for "getting motivated" that is unlike all others you've likely seen before. And you'll finally get the truth about neuroplasticity —the rewiring of neural circuits that underlies all forms of learning (and forgetting), and that is often referred to simply as plasticity. Most everyone has heard of plasticity but honestly, most of the ways I've seen it discussed publicly are flat-out wrong. Perhaps you've heard that "Every experience you have changes your brain." That is not true.

Or "It takes twenty-one days to form a new habit." Wrong again. It can take a few days or a year, based on your level of motivation. Perhaps you've heard that "Your brain makes new neurons for your entire life." (The oficial terrn for that is neurogenesis.) Alas, barely any new neurons are added to the brain after puberty except for in your olfactory system, which allows you to smell, and those are mostly to replace neurons you've Iost. Starting to get depressed? Well don't, because despite these myths, we all possess a tremendous ability to rewire our brains and to learn at any age.

The key is to understand the type of plasticity you're after. What we're talking about here—the rewiring of our brains that occurs as part of

learning at any age—is a process that involves a specific set of steps that must be completed in a specific sequencefor learning to occur. I call these steps self-directed adaptive plasticity (SDAP). Mastering these steps opens the doors to learning pretty much anything you set your mind to. And that's what this chapter is about.

The Steps and Sequence of SDAP Let's start with an overview of how SDAP occurs: Get alert. SDAP requires you to focus on the material at hand. Alertness sets the stage for focus; it's a precursor. 1'11 explain all this in the pages ahead. So get alert and.. Get and stay focused.

Focused on what? On implementing a specific strategy during your learning bout. As yougo... Maintain a state of urgency. Your internal state really matters as you learn.

You can't be too relaxed, but you can't be stressed either. I define a state of urgency as simply a state of persistent motivation. It's the state to aim for because it has you stay alert but not too alert and.. Embrace errors. Errors that arise from genuine attempts to do something are key signals for triggering plasticity.

Your nervous system is stimulated to rewire largely based on what you fail to do, not what you do correctly. Rest. Sleep is when neural rewiring actually occurs, especially during REM sleep the first night after a learning bout. There are other ways you can incorporate rest—like non-sleep deep rest (NSDR) and short breaks—to improve performance. Repeat.

Consolidate newly learned material or skills through repeat performance, either through physical repetition or taking yourself through the material or skill using your imagination (seriously). Mental self-testing is especially effective for reinforcing newly formed brain connections.

Each of these Six steps is valuable in its own right, and can be applied to various areas of life. Thus, although this chapter is framed in the context of plasticity, each of the protocols has other important applications too. 1'11 explain more as we go. Types of Neuroplasticity Before we explore each step of SDAP, let's get Clear on what we're talking about when we talk about plasticity. All sustained behavioral and cognitive change is the result of plasticity. These changes come in two broad types: adaptive and maladaptive.

Adaptive changes are desirable—this is the plasticity that accompanies learning a new language or a new artistic or athletic skill. Maladaptive changes are the ones that make life harder. Consider when a brain injury renders someone unable to produce coherent speech, or a traumatic event

leaves someone stricken with PTSD. That's Still plasticity but obviously not the sort you want. These two types of plasticity happen in different ways.

Maladaptive plasticity, as in the brain injury I described above, tends to occur fast and is hard to undo. Adaptive plasticity, like learning that new language, takes more effort, and reverts somewhat easily, at least at first. I realize that's not convenient but the sooner we accept that reality, the further along we can all be in our efforts to learn new things. Understanding Why and how maladaptive brain changes occur Will help illustrate how to get the adaptive brain changes that you want, and make them happen more quickly. Psychologically traumatic events trigger astonishingly fast and stable changes to nervous system wiringu This rewiring happens not just in the brain but in the patterns of brain-to-body and body-to-brain connections, and hormone-control pathways.— The duration of a traumatic event can be very brief—no more than a few seconds— yet bring about persistent or recurring anxiety, sleep disturbances, and physical challenges, like chronic pain,3 rashes,4 gut issues,5 and more, that last for years.

This is essentially one trial learning. Something happens and we are (seemingly) forever changed. The evolutionary logic is straightforward: One trial learning from traumatic events helps ensure we avoid similar circumstances in the future. It keeps us safe. How one trial learning happens brings us back to an analogy discussed in chapter 3.

Remember the autonomic seesaw? A stress-inducing event or thought causes your autonomic nervous system to increase your sympathetic tone, which makes you alert. That sharpens your focus. If sympathetic tone is too high, however, it feels like anxiety or panic. When we experience a traumatic event, our brains quickly trigger the release of a lot of adrenaline into our bodies and a lot of norepinephrine in our brains.6 Together, these two neuromodulators create all the necessary conditions for plasticity to occur.

We're not talking about a bit of normal life stress. We're talking about something you perceive as really scary that causes a large and rapid shift in the neurochemical milieu of the body and brain.

that what is happening is unique and important, and that they need to change to become better equipped for similar circumstances in the future. Another useful way to think about it is that your brain must get the message that there is a gap between its current configuration and the demands you're facing—a deficit of sorts that it needs to overcome. The deficit is resolved by making new brain connections. Now of course your

neurons don't talk to each other in words. They don't say, "This is important, and we have a deficit between where we're at and where we need to be, so we need to change." Neurons communicate in the language of chemicals and electricity.

So this message is communicated electrochemically, as a shift in your brain and body's neurochemical milieu that in turn alters the amount of firing in the brain circuits that need to change for the desired learning to occur. And when it comes to shifting neurochemical milieus—that is largely the job of neuromodulators. The neuromodulators dopamine, serotonin, adrenaline, norepinephrine, and acetylcholine each serve different purposes, depending on where they are released in your brain or bodyg It's incorrect to say that dopamine alone drives motivation, for instance, or that serotonin controls mood. It's far more varied and nuanced than that. In terms of plasticity, all these neuromodulators can alter the level of electrochemical activity in the circuits I just described.

In other words, these neuromodulators can change the amount and the volume that neurons "talk" to each other.

In fact, in the context of plasticity, you can think of a neuromodulator as a microphone lowered over a specific conversation in a crowded stadium. Like a stadium on game day, brains are noisy. They are busy carrying out all kinds of everyday electrochemical duties. The "microphone" makes a conversation louder against the background noise. This is one common way plasticity occurs.

The faster and greater the shift in the activity of a given circuit relative to its previous levels, and relative to the background (more electrochemically active), the more likely that circuit Will alter its structure and connections—in other words, the more likely plasticity of the circuit Will occur. It's also true that if neural circuits are quieter (less electrochemically active) against the chatter of usual background brain activity, they can weaken. Different contexts spike the release of different neuromodulators alone or in combination, which amplifies the conversation of the neural circuits active in that context. For example, dopamine is a major neuromodulator released in response to novelty and surprise, which facilitates plasticity for learning and memory.2 And acetylcholine is released at high levels under conditions of heightened focus, but not all over the brain; rather, it's released specifically at the sites in neural circuits that are active. Regardless of which neuromodulator or combination of neuromodulators is released, what happens next is largely the same.

Since the brain is homeostatic—that is, it seeks balance— rather than maintain high levels of activity, the neurons that are part of a given conversation modify themselves so they become more easily active should that context occur again. They may add more receptors to render themselves capable of capturing more neurotransmitters or they may sprout a tiny neurite, an axonal (output) or dendritic (input) branch that becomes part of a new set of synaptic connections with other nearby neuronsn All this changes the threshold for activation of that circuit going forward, and on and on.J2 Voila! That is learningdriven plasticity. Other forms of plasticity involve similar cellular and molecular changes but the conditions that create them are different. SDAP is about creating this situation of specific neural circuits rewiring in specific ways on purpose.

The steps involved in this process are designed to help you do just that. Where to Begin Above, I noted that the first step of SDAP is getting alert and the second is getting focused. These two work together in a positive feedback loop. The cool thing is even the mere act of trying to focus shifts us into a slightly more elevated sympathetic tone, which causes our levels of alertness to increase, which then causes activity in the circuits for focusing to increase, which further causes levels of alertness to, you guessed it, increase. For those who struggle with focus, take solace in the fact that the circuits involved with focus are amenable to plasticity too.

As you move through the SDAP process described in this chapter, and practice the various protocols here, you'll learn new things, but you'll also get better at focusing, which Will help you to learn more new things, and so on.

There are multiple ways to create the increase in alertness and focus needed for SDAP. 1'11 cover all your options—behavioral, supplement based, pharmacologic—below. But remember this: Alertness and focus are necessary but not suficient for plasticity and learning. To trigger learning, you're also going to have to make errors from genuine attempts, what I refer to as repetitions, or reps. Not deliberate errors, and not half-baked efforts, but rather, errors that emerge from truly exerting yourself to learn new information, new movements, even new modes of thinking. Consider this: If your nervous system can already generate the activity patterns that enable you to do a certain task— perform a dance step, for example, or order coffee in a foreign language—why would it change?

It doesn't need to! You're already equipped with the neural circuits to accomplish that task. You need to try, and fail, and try again. This is tough to accept, especially for perfectionists, but all who embrace it thrive.

Especially those who come to recognize limbic friction, the slight agitation that accompanies effective learning, as a state to seek, not Shy away from.

The importance of making mistakes and trying again doesn't mean you have to toil endlessly. SDAP can be enjoyable. You'll likely experience some limbic friction but guess what? Once you learn to expect it, and you realize that it helps not hinders your ability to learn, you too Will be seeking it out. You may be wondering "Why all these conditions for plasticity?

Isn't there a plasticity pill?" No. As I mentioned earlier in this introduction, plasticity is energetically expensive.

sequence is more important than any one tool listed here. This Will be an important reframing for a lot of people. I mentioned that there wasn't a pill for plasticity, but people love to think about what they can take to increase neuromodulator levels to heighten focus. Adderall, energy drinks, nicotine, modafinil, microdoses of psilocybin, et cetera, are used by many people working in business, tech, sports performance, finance circles, and beyond to try to focus and learn better. The use of such stimulants is widespread on college campuses and among high school students in the US,J-3 and millions of adults use them daily too.

Their widespread use merely speaks to how hard most people find it to focus intensely enough and push through limbic friction. I'm not judging anyone. For example, I, like most adults in the US,J4 use caffeine daily to increase focus. I occasionally rely on other pharmacologic tools for focus too. In this chapter, 1'11 cover the popular as well as the rarely discussed ones.

That all being said, the main focus here is behavioral tools. They may not seem as sexy, but many rival the pharmacologic tools. It truly doesn't matter what you use to get alert and focused, as long as it's safe. If you opt to use nicotine or modafinil or meditation or good 01' sleep and hydration, those obviously have different safety profiles. You can use white noise or various sound tones.J5 These all have data to support them in the context of focus and learning.

You can cycle different tools in and out, by the day or by the month. The point is: You have choices. Think of them as tools in your SDAP arsenal. Stick with It We know a lot more about plasticity now than we did twenty-five or even just five years ago. Prior to the 2000s the widely held belief—even among neuroscience researchers—was that the opportunity for adaptive plasticity mostly closes after age twenty-five.L6 That idea is now dead.

We know people can rewire their brains and learn throughout their lifespans.UZ Also mostly dead is the idea that learning-associated plasticity involves the addition of new neurons. Plasticity in humans is almost exclusively the consequence of changes to existing neural circuits, not adding new neurons to those circuits.J-g And yet, the results you can achieve when you incorporate the best of what we know about SDAP can be extraordinary. Here is an overview of SDAP, once more: • Get alert. • Get and stay focused. • Maintain a state of urgency. • Embrace errors that arise from genuine attempts. • Rest. • Repeat your performance. I've listed these again because the steps and the sequence are non-negotiable. Not because I say so, but because that's how our neuroplasticity works.

There are things you can do to accelerate the process: all the tools I mentioned above (and more), along with micro-pauses, post-learningbout NSDR, adjusted sleep protocols, and hypnosis. I cover all these in the pages that follow. I began this chapter with this basic process because it can be applied to learning anything. Similar to exercise, SDAP is not easy. But with time, each step becomes familiar.

If you think of SDAP this way—as a process that consistently leads to a desired end—it can become a pleasure.

LEARNING PROTOCOL Shift into a State of Alertness and Focus

WHAT TO DO

• Getting alert and focused is the first step in SDAP. You have to be in a state of elevated autonomic (sympathetic) arousal, but not stressed. • Use tools to help you achieve this state. Such tOOlS are many and varied, and fall into three general categories: • Foundational • Behavioral • Pharmacological • Put your Phone away. You may think you have an issue with getting focused, but it's likely that you are distracted by something in your surroundings—the smartphone being the major culprit. Studies show that even just having your Phone in the same room in a bag behind or under your chair reduces your ability to focus on cognitive tasks and/or increases the energy that it takes.12

HOW IT WORKS

You might be thinking "Why would anyone need a protocol for focus?"

oversized motorcycle helmet designed to be worn while seated at a desk, created by industrial design company Hochu rayu.24 The rationale was the same, all that time later: Block out peripheral vision and outside noise to

help people focus. The Helmfon, as you might imagine, also did not gain traction. Even if it had, the advent of social media and other online activities essentially moved the outside noise of the entire world directly into one's central field of view. So much for helmets. Other long-used tools include nonamphetamine stimulants such as caffeine, which blocks adenosine receptors in the brain that would otherwise trigger sleepiness, and nicotine, which triggers the release of several neuromodulators discussed in the introduction to this chapter.

The use of caffeine in the form of tea or coffee dates back more than two thousand years. But coffee and tea didn't become popular in Europe until the seventeenth century, the result of invasions and trade.25 Today, approximately 90 percent of the world's adults—that's estimated as more than five billion people (!)—drink caffeine every single day. Later 1'11 explain how to use caffeine specifically to enhance focus, but meanwhile those stats reveal that caffeine is the most popular drug on the planet by a huge margin. (If you're wondering, alcohol and nicotine are in a distant second and third place, respectively.) Nicotine, mostly in oral forms like pouches, gum, and mints, has made a big comeback as a focus aid in recent years. I recently gave a talk to several thousand undergraduate men and women at an East Coast university and asked the auditorium for a show of hands of "who in here takes nicotine daily?" About 30 percent of them raised their hands. (In a more private setting— a smaller group of about fifty male students— about two-thirds said they use nicotine two or more times per day every day.) This is obviously not a controlled study, nor did I count each hand, but even if these are artificially high estimates, it's Still pretty shocking. Or is it?

Nicotine is an unusual stimulant because it makes people alert but calm—a highly desirable state for learning. Unless smoked, vaped, dipped, or snuffed, nicotine does not cause cancer, but it is highly addictive and it raises blood pressure.26 Later, 1'11 explain how, assuming you're Okay with the risks, nicotine can be used to enhance focus. There are a few safer alternatives, some of which work through the same mechanisms. And there are the amphetamines. First synthesized in Romania in the late 1880s, amphetamines are compounds that increase levels of some of the key neuromodulators discussed in the introduction to this chapter, like norepinephrine and dopamine.ZZ These stimulants were initially marketed in the 1930s as a treatment for narcolepsy, but soon hit the college scene when research showed a link with increased scores on intelligence tests.

They were also investigated for the treatment of attention deficit hyperactivity disorder (ADHD) and as appetite suppressants starting even

back then.22 A decade later, both German and Allied forces used amphetamines during World War II to stay awake and alert.30 Recreational use of amphetamines exploded in general popularity starting in the 1950s, partly due to a surplus from World War II. By the 1960s, there was an epidemic of amphetamine dependency and abuse in the United Statesn They were made illegal for non-prescription use in the early 1970s, classified in the same way as cocaine and morphine, and their use dropped. That is, until the FDA approved amphetamines for the treatment of ADHD in 1996.32 Drugs marketed for ADHD like Adderall, Vyvanse, et cetera, are amphetamines, and they work quite well to increase alertness and focus. Current exact rates are hard to come by, but it seems that up to 35 percent of college students use stimulants at some point during their undergraduate years.33 For people without a genuine clinical need for amphetamines, their use can be problematic. They are effective for helping people complete work that relies on skills they already have, but far less effective for learning new skills.

They are lousy for contemplative thinking, are very addictive. That's too, and creative endeavors. And they habit forming, and in many cases the history. And while getting and staying focused has always been hard, it seems it's now harder than ever. The good news is that accessing a state of focus necessary for SDAP is possible for anyone.

It starts with understanding the states of mind and body you're after. You'll recall that negative experiences automatically set off the learning process. No coffee, no tutor, no motivational video or mantra required. The fast, massive increase in sympathetic tone caused by adrenaline and norepinephrine ramps up activity in relevant brain circuits. It's proof that if the right electrochemical systems get triggered, we all have the capacity to focus.

Again, I'm It's where one person places their hands out, palms up, and you put your hands on theirs, palms down. The person with palms up flips their hands over to try and tap the top of your hands before you can pull away. You don't know when they Will make their move. You're alert and focused, anticipating something Will happen. Learn to recognize that state.

That's the ideal level of autonomic arousal for a learning bout. With a bit of practice you'll be able to sense when your autonomic arousal is climbing too high or dropping too low, and keep yourself in that ideal state. In other words, you learn to get better at entering the learning process. Is Being Focused the Same as Being in a Flow State? No.

Don't confuse this state with a flow state. In flow states we forget the rest of the world (and even ourselves), and we do things as if on autopilot.32 Flow states are ideal for task performance—executing cognitive and motor skills we have already mastered. We don't learn in flow states, we implement what we already learned. The person who first described flow states—the Hungarian psychologist Mihaly Csikszentmihalyi —accurately described this feeling as when the challenge of a task is matched with your existing skill level, not too easy, nor too hard.. just challenging enough to engage you and "an absolute sense of control."40 That last part—"an absolute sense of control"—means your brain already has the neural circuits in place that it needs to execute the task. And as we discussed above, that means no rewiring... no plasticity is triggered, and no learning occurs.

To trigger plasticity and start learning, you must push past your current abilities. You have to get stumped, miss the mark, and most importantly, you have to feel the agitation of making errors. Recognize that as limbic friction, and seek it out. Ask any elite performer how they got so good at their craft and they Will tell you, "Practice, lots of practice." But there are two kinds of practice. There are the hours upon hours spent drilling what we already know how to do and there is the practice of pushing ourselves to learn new skills.

The latter is accomplished by making errors through genuine attempts. As the great Twyla Tharp, world-renowned choreographer and dancer, put it, "The more you fail in private, the less you Will fail in public."41 Twyla understands that elite performance requires genuine attempts —failing and failing again (and again), until the gap between one's goal and actual performance is reduced to zero. The Learning State The experiments that perhaps best illustrate the vital role of focused attention in learning come from the work of neuroscientist Mike Merzenich, PhD, and his lab at University of California, San Francisco.42 The studies were done in the 1990s using monkeys and later adapted to studies in humans. The monkeys were trained to place their fingers on a rotating cylinder that had both coarse and fine bumps laid out in rows across its surface.

The monkeys quickly learned that if they detected a difference between the coarse bumps and the fine bumps, they earned a reward. To signal when they noticed that difference, they used their other hand to move a joystick. Over time, the researchers made the task harder by making the difference in the bumps progressively smaller, which required more focus from the monkeys. Not only did the monkeys learn to perform the harder task, but the researchers observed significant changes in the brain areas

representing sensations in the fingers that were touching the cylinder. In other words, they observed plasticity associated with that learning.

These were adult monkeys, too, showing that plasticity was not just confined to developmental critical periods as had long been believed. That was interesting but it was just the beginning of the story. In other studies from Merzenich, monkeys had to learn to discriminate between different sound tones. In that case the regions of the brain responsible for hearing changed. Perhaps not surprising, except that the monkeys were experiencing bumps on their fingers at the same time they heard the tones.

So Why did only the auditory areas of the brain change? Because the monkeys received rewards only for learning sound tones, so that's what they focused on. Together, these two sets of experiments showed that mere experience doesn't cause neuroplasticity. Learning-associated brain plasticity only occurs in the circuits associated with where our attention is placed. Since those landmark studies, there have been thousands of experiments exploring the participants were instructed to pay attention to.

The finding leads to a profound insight: that the plasticity they experienced was tied to the act of looking for something and not solely to the success offinding that something. As the authors concluded, "Learning Will occur only in those neurons that are activated by the stimulus and that are at the same time relevant for the task. Which neurons are relevant [for neuroplasticity] is dictated by the mechanism or strategy used for task performance (and not merely a stimulusresponse correlation)." These results, along with the earlier studies mentioned here, point to the crucial role of focus in plasticity and learning. Which brings us to the common saying "fire together, wire together."42 People often use this phrase to capture the idea that habits form from neurons being activated at nearly the exact same time. The thinking goes that neurons firing simultaneously or one after the other causes them to become wired together, which Will lead them to readily fire together in the future.

But the term "fire together, wire together" is actually unrelated to the kind of learning we are talking about in this chapter—achieving new skills, developing new behavioral patterns, et cetera. This is not a trivial point. This phrase is often attached to the false notion that people can learn when in passive or semi-engaged states. It implies that we can simply expose kids to music to enrich their brain. Not sos It also implies that simply reading or doing something repeatedly is enough to create new brain patterns.

Alas, not so. Oh and the "fire together, wire together" phrase is constantly misattributed to psychologist Donald Hebb when in fact it was Carla Shatz, the developmental neurobiologist, who said it first— in the context of developmental brain wiring, not learning.50 The point is this: Whether young or Old, ifwe want to learn, we need to focus intensely on new information, or in the case of cognitive or motor skills, on our attempt to perform them. The good news is that with the right tools and a little practice, anyone can access focus... and learn. Tools for Focus As I mentioned in the introduction to this protocol, we humans have long sought tools to make this endeavor easier. Here's where science puts us today.

There are three categories of tools to get alert and focused. Foundational Behavioral Pharmacological The tools offered here can be used for learning bouts, as described above, or even for what I call work sessions—time when you're not necessarily tackling new information or skills, but grappling with some work or physical challenge that simply requires intense focus. FOUNDATIONAL TOOLS FOR FOCUS These tools are the bedrock upon which all the other suggestions in this protocol depend. Without them, the rest of the tools in this protocol Will be far less effective. Get the foundational tools right, and you'll greatly amplify the potency of all others you opt to use.

Foundational Tool 1: Sleep In chapter 1 when I said that sleep is the foundation of mental health, physical health, and performance, that included learning of all kinds. When we do not get enough quality sleep, or if our sleep timing is altered by more than an hour or two from our normal schedule, our levels of alertness get harder to regulate and so focusing becomes an even bigger challenge. Thus, aim to get adequate sleep using the protocols in chapter 1. (See Sleep Protocols 1—6 for a reminder on the basics.) Foundational Tool 2: NSDR If you've had less than adequate sleep and you're trying to lock in on a learning bout, consider doing ten, twenty, or, if possible, thirty minutes of non-sleep deep rest (NSDR) at some point prior to starting that learning bout. NSDR, while not a replacement for sleep, is your best bet to restore mental and physical vigor if you didn't get enough sleep. As mentioned in Sleep Protocol 8, NSDR is preferable to naps, because it doesn't create sleep inertia so you won't finish a session feeling groggy, nor does it disrupt your normal sleep schedule.

A short nap of thirty minutes or less can also restore mental and physical vigor, but midday napping for more than thirty minutes can really mess up your nighttime sleep. Refer to Sleep Protocol 7 for best practices. Foundational Tool 3: Stress Control The other key to an effective learning

bout is to make sure you don't show up in a state of excessively high sympathetic tone, or what we commonly call stress. Stressing about something other than what you're trying to learn steals your focus. If you're worrying about your to-do list, it's hard to focus on learning a new song on the guitar.

And stressing about what you're trying to learn prevents you from staying in the alert but calm state you're after. Chapter 3 provides tools you can use when you aren't actively engaged in a learning bout—as well as tools you can use during a learning bout, such as the physiological sigh. Both types of tools are effective to help maintain your level of arousal near the peak of the inverted U-shape curve that is ideal for learning. Foundational Tool 4: Nutrition If you're even mildly dehydrated, your focus won't be at its best.5-J Drink water and, if you need to, add some electrolytes, which aids the rehydration process. The same goes for your blood sugar levels—make sure yours are relatively stable by not reaching for sugary foods when you know you want to be focused.

Chapter 4 described how to eat to fuel your body for all types of activity, and that includes learning bouts and work sessions. Those protocols offered guidelines, but there are options for how you adapt them for your schedule when it comes to learning. For example, in the first two to three hours of my day, I focus best if I'm fasted. I make sure I'm hydrated, and that I'm caffeinated, too, but unless I wake up especially hungry (a rarity for me), I don't eat yet. If I already exercised that morning, however, I eat, and then I start work.

can significantly improve your ability to focus. Some assist the transition into a state of focus, and others help you stay there. Behavioral Tool 1: Meditation Meditation can take many forms. One shown to improve focus is as follows: Set aside about ten to twenty minutes, sit or lie down, close your eyes, and direct your mental attention to the location between your eyes, or concentrate on your breathing, or both. As you strive to maintain those points of focus, thoughts, sensations, and outside noises inevitably distract you.

Keep redirecting your attention back to the location between your eyes and/or breathing, over and over again, as many times as needed until the allotted time elapses. From novice to expert, no one is expected to maintain perfect focus for the whole session. It's the redirecting of your attention that is the key component. It's analogous to the final few repetitions of a weight lifting set. It's what drives the nervous system changes that make it easier to focus the next time you meditate, or in any

other activity.

It's training; it's what is meant by the phrase "meditation is a practice." A study by researchers at New York University found that meditating in this way for just thirteen minutes per day improved participants' ability to deliberately focus and to regulate their emotions, two strong indications that this behavioral tool can enhance and speed up learning that requires concentration.54 Shorter meditations of five minutes per day have also been shown to help.55 You may be thinking this seems like the micro meditation described in Stress Control Protocol 4, or any other meditation... and you would be right. All meditation is perceptual exercise where you intentionally limit what you pay attention to. Remember the Isolator? That was a perceptual filter designed to restrict your visual field—that is, intentionally limit what you pay attention to. Knowing that, it should make sense that what you elect to focus on during a meditation is less important than the act of directing, and redirecting, your attention back to the same place.

You can close your eyes and focus on the top of your head, your feet, your lower belly, and your breathing, or you can open your eyes and focus on a point on a wall in front of you (allow yourself to blink as needed, of course). Studies show improvements in focus stem from all such meditations, because meditation is a perceptual exercise for improving focus.56 It just so happens that it also helps reduce stress, boost mood,5.3 and improve sleep.5 A recent study even showed that meditation can induce a similar glymphatic washout process—that is, the removal of metabolic waste from the brain —as observed during sleep—albeit to a lesser degree.60 With a year or more of experience, meditation can allow for profound insights into how our thoughts arise (and pass) and for some people it can be transformative for understanding consciousness, but those are not the immediate benefits. Nor should anyone wanting to improve focus Shy away from meditation thinking they have to experience those sorts of major insights for it to "work." Meditation done simply as a perceptual exercise is powerful. And the effects show up fast: less procrastination, less impulsivity, and better learning. Ifyou are very limited on time, here's a simple ten-breath meditation to try immediately upon waking, while Still lying in bed: Keep your eyes closed and focus on your breathing for ten Slow breaths.

If your mind wanders before you reach ten, restart at one, and do that as many times as needed until you reach ten focused breaths. Then get up. After a few days, you'll notice improvement. Of course, five minutes, as described in Stress Control Protocol 4, and eventually ten, fifteen, or

twenty minutes of meditation would be even more beneficial. I've heard from many people who do this regularly—meaning three to seven times per week—that they can focus better, and redirect their focus if it wanders.

You'll experience the other positive effects of meditation explained above, such as stress reduction and improved sleep too. My lab found that just five minutes a day of meditation provided these benefits.6-J What time of day or night you meditate is far less important than if you do it regularly. In fact, there is no bad time to meditate as long as you're in a safe location. If your back or knees ache from sitting cross-legged on the floor, remember that you don't have to do that. Make yourself comfortable.

Sit in a chair or lie down in bed. In my experience, it is the rare person who adheres to a daily meditation practice consistently over many years. Most become the "I used to meditate every day, I need to get back to it" people, not unlike the "I used to work out every day, I need to get back to it" people. I believe that happens because we shoot for ten to Clinical hypnosis is supported by solid research as a tool to treat chronic pain, and to help with smoking cessation, sleep challenges, anxiety, even gut health issues.é2 Through decades of research, including brain imaging and randomized controlled trials, Stanford University psychiatrist David Spiegel and his lab discovered that clinical hypnosis stimulates specific types of plasticity that allow for better autonomic selfregulation.6 For example, hypnosis appears to reduce activity in the part of the brain called the dorsal anterior cingulate cortex, indicating diminished self-monitoring. An accompanying increase in connection between the dorsolateral prefrontal cortex and the insula indicates that the brain is doing a better job of regulating the body.

These and other findings point to a state in which people are simultaneously less self-conscious and yet better at self-regulating. Hypnosis won't allow you to immediately speak a new language or perform the perfect golf swing, but it is very effective for teaching people how to access otherwise inaccessible brain and bodily states, such that they can more easily reaccess those states once hypnosis is over. For example, someone with an intense fear of public speaking can experience being alert but calm while onstage in front of a big audience. With just a few sessions and the occasional brush-up session, people can get very good at this kind of autonomic self-regulation. Hypnosis, in other words, is about learning how to set your state of mind.

If you have trouble with focus, or any of the other issues mentioned above, you may consider hiring a licensed clinical hypnotist to work with in person

or online. Alternatively, some years back, Dr. Spiegel developed an app called Reveri that provides this tool to all (I'm an advisor for Reveri). It has you listen to a brief recorded hypnosis script (in Dr. Spiegel's voice) on your phone, a few times per week.

Reveri has hypnosis scripts for improving focus, and also for improving sleep, reducing chronic pain, and much more. Each takes about eight minutes to complete. If you try Reveri, remember this: It's not magic. It's also not mind control and it's most definitely not stage hypnosis. It is a tool for teaching your brain how to access different states, including focus.

Behavioral Tool 3: 40 Hertz Binaural Beats, 20 Hertz Oscillatory Induction, and White, Brown, and Pink Noise The use of sound as a tool for focus is all about entrainment, or the matching of our internal rhythms to an external sensory event—in these cases, the frequency of sound waves. You may remember the term entrainment from chapter 1 and chapter 5. There, I talked about entraining your circadian rhythm to light, for example, by viewing bright light in the first hour after waking up. There are also ways to entrain to events in the outside world. For example, when you exercise at roughly the same time of day, you experience an increase in catecholamines and certain hormones at that time.

Then after about three days or so, you automatically start to feel an increase in energy just prior to when you usually exercise. Entrainment works based on your nervous system's expectation of certain things happening.

Sound waves can entrain our brain, too, and they do so very quickly. When we hear certain frequencies or rhythms, our brain starts to entrain to the predictable features in the sound by our neurons firing in regular fashion. Scientists have figured out how to leverage that and have created specific sound patterns that increase brain activity in areas that underlie focused states. One example is binaural beats. Two distinct frequencies of sounds are played in each ear.

When that happens, a cluster of neurons in your brainstem called the medial superior olivary nucleus performs a subtraction of the two frequencies. So if a 400 hertz (Hz) sound arrives at one ear and 440 Hz at the other ear, it leads you to perceive a 40 Hz beat. That sound frequency, in particular, has been shown to improve levels of focus, and even certain forms of memory.6 The precise mechanisms for this are Still being parsed but there is evidence that 40 Hz sound can do this in part by causing release of acetylcholine from the medial superior olivary nucleus and the release of dopamine in the striatum, a brain area involved in movement

and motivated attention. Some evidence even shows that 40 Hz sounds can trigger the brain circuits involved in focus and memory to enter what are called gamma oscillation states, a pattern of neural activity that is favorable for learning.é6 (Interestingly, there is parallel evidence that activating these circuits using clinical devices may also help offset some of the early symptoms of age-related cognitive decline in Alzheimer's disease.6-Z) Not everyone appears to benefit from this tool however. It seems if people already have elevated levels of these neuromodulators prior to the sound exposure, the benefits of binaural beats are less apparent or even absent.é-g Alas, without doing an fMRI and blood draws, there is no way to know your baseline dopamine and acetylcholine levels.

Considering 40 Hz binaural beats are safe and cost nothing (there are dozens of free apps, videos, and audio recordings of these on Apple music, Spotify, and YouTube), Why not try them and see? You have nothing to lose and perhaps something significant to gain in your efforts to focus better. Here's how: Try listening to 40 Hz binaural beats for five minutes prior to a learning bout or work session. Once you determine if that works for you, try listening to the beats continuously throughout. (Of course, if you are trying to learn something that requires you to listen to instructions, you'll only be able to use the 40 Hz tones prior to the learning bout.) Also, note that it's recommended you use headphones for binaural beats as it's the best way to entrain your olivary nucleus and other brain circuits to 40 Hz gamma frequencies. There are exciting new data suggesting that sounds of about 12 to 20 Hz—which cause the brain to entrain to beta, not gamma, waves—may also improve concentration and reduce mental fatigue" Music with 20 Hz as the dominant frequency may be especially effective here.

I personally have tried 40 Hz binaural beats, using one of the many free binaural beat apps out there (I like BrainWave). I've also used white and Pink noise, both of which have been shown sessions, so go figure. Be a scientist of yourself, and customize your own protocols for focus. One thing is certain: Sound can be a powerful tool for entraining brain states required for learning. A note before we continue: Throughout this chapter I've been highlighting the important role that adrenaline and norepinephrine play in focus, and the formation of memories and learning.

Soon we Will cover ways to boost these and some other closely related neuromodulators, like dopamine, using pharmacology. But first, there are established behavioral tools to significantly and safely increase adrenaline and norepinephrine without taking anything. These won't surprise you because I covered them earlier, in different contexts. Behavioral Tool 5: Physical Exercise Physical exercise causes meaningful increases in

adrenaline through now-well-understood neural circuits, stemming from your motor cortex in your brain all the way down to your adrenal glands near your lower back.Z3 Any vigorous physical activity that involves the large musculature of your legs and core (abdominals, lower back, etc.), Will trigger the release of adrenaline, in a manner that scales with the intensity of the activity. Think: compound lower body exercises like squats or deadlifts, running, martial arts, tennis, and on and on.

Adrenaline does not cross the blood-brain barrier, but it binds to receptors on your vagus nerve—the twoway neural superhighway that bridges your body and brain.Z4 That, in turn, stimulates the release of norepinephrine from your locus coeruleus (remember that, from chapter 1?) and other brain 7.5 areas involved in generating a state of focus. That's one mechanism, but there are others at play, including increased blood flow and brain oxygenation, and lactate-induced elevations in brain-derived neurotrophic factor (BDNF) (which you'll recall from chapter 2… see how all of this comes together?). The effects of all this on your focus and cognition are significant, and can last several hours afterward. One of the best ways to use exercise as a tool for focus is brief high-intensity interval training (HIIT)—for example, after a five- to ten-minute warm-up, I like to do a thirty-second near-allout sprint, followed by thirty seconds of rest, and again, for a total of eight sprint-rest cycles. Other HIIT options work too.

You can find dozens of ideas for these online with a simple web search. It's crucial to warm up for a few minutes before the first round and important too that you select a form of exercise that you can perform without getting injured. I like to run outside on a track or use a stationary assault bike, but you may prefer to swim, use a rower, a stair climber, et cetera. There are also studies showing improved focus and cognition after sustained twenty to fortyfive minute cardio training sessions at lower intensity (60—75 percent of maximum heart rate), although the effects of this on focus appear more 7.6 transient compared to more intense exercise. That makes sense, given HIIT elevates adrenaline and norepinephrine more than lower intensity zz steady state cardio.

Fewer studies have explored how focus might be increased after resistance training. But assuming the sessions are not overly long or intense, I would expect them to impart a similar focus-enhancing effect as cardio training sessions (see chapter 2 for more on the ideal duration and intensity for those sessions). Experiment to find the form of exercise you like best to enhance focus. Make sure to time the duration and adjust the intensity of effort so it doesn't exhaust you. Behavioral Tool 6: Proper Breathing Sleep apnea is a condition where people under-breathe during sleep.

Nasal obstructions, excessive weight, and certain medications cause it. If you snore, there's a good chance you have sleep apnea. The result? Your brain is deprived of oxygen at night. Sleep apnea is a dangerous condition, because the health impacts creep up over time—causing fatigue, cognitive impairments, heart issues, and more" Some of these can be reversed with tools to correct breathing while you sleep, but longlasting impairments can occur.

Why am I discussing sleep apnea in a protocol on focus? Because many people have daytime apnea that impairs their focus too.Z2 Daytime apnea is when you hold your breath while working. Chronic mouth breathers experience the same. When your brain doesn't get enough oxygen, you get tired and you can't focus. To feel alert and focus intensely, you need to fix sleep and any daytime apnea.

Luckily, it's easy to do. First, make sure you're breathing through your nose anytime you don't have to speak, eat, or exercise hard. Second, learn to breathe with your diaphragm, also known as belly breathing, in which your belly expands as you inhale. This approach increases the amount of oxygen in the blood and promotes a more relaxed state. You'll be amazed what a difference that Will make for your energy levels and focus.

Check right now. If you're mouth breathing, switch to nasal breathing. Does your belly move out as you inhale and move in as you exhale through your nose? If so, great, you're nasal diaphragmatic breathing. If you're having trouble, imagine inhaling through your nose as if trying to direct the air backward—in other words, think about drawing the air "in and back" through your nose at an angle parallel to the roof of your mouth, as opposed to inhaling "up" your nostrils toward the top of your head.

I know that sounds weird, but try it. You'll notice that you immediately shift. And no, you don't have to think about your breathing all day. After two to three breaths done consciously that way, it Will carry on reflexively. Nasal diaphragmatic breathing in the daytime also helps you default to the right breathing pattern when you sleep, which can help offset sleep apnea, so it's a twofer.

A few other issues with breathing to address here. If you have a badly deviated septum or challenged nasal breathing brought about by cartilage growth in your nasal passages, you need to fix that. The simplest way to open your nasal passages is to, again, practice nasal diaphragmatic breathing. Your sinuses are not fixed, but over time this effort Will help open them up. Extreme cases of nasal passage collapse or other blockage may require surgery.

People can often achieve some results from wearing a magnetic nose bridge for a few hours a day and/or at night.

Behavioral Tool 7: Cyclic Hyperventilation Cyclic hyperventilation might sound familiar, but there's new context for this powerful tool. It's quite simple. You hyperventilate. Intentionally. Not a bunch of quick shallow breaths—that's panting!

Instead do ten to twenty cycles of a big deep inhale through your nose, using diaphragmatic breathing as described in the section above, and exhale, letting the air passively "fall out" from your mouth. You don't have to heave into the inhale to get the effect you're seeking. Work up to it gradually, and focus on long, diaphragmatic inhales through the nose. Increase the vigor after a few repetitions. After ten to twenty breaths this way, you Will reliably feel much more alert.

As covered in chapter 3, you can spike your levels of adrenaline and norepinephrine and dopamine—the so-called catecholamines— and thereby increase your alertness and focus, by breathing in this specific way. It works by causing increased discard of carbon dioxide, which is called hypocapnia, and increased alkalinity of your blood pH, which is called alkalosis. That in turn triggers fast and significant increases in adrenaline and norepinephrine release in your body and brain. If you think breathing to shift catecholamine levels is not very potent, think again. This tool works fast and extremely well to increase alertness, so you can focus on whatever task lies before you.

It's a causal relationship. In experiments where people hyperventilated through an air mask with artificially elevated C02, their epinephrine and norepinephrine levels did not increase, nor did they achieve alkalosis. This tells us it's not just the mechanics of deep breathing, it's about increased discard of C02. By the way, that is Why people having a panic attack are told to breathe into a paper bag; it's to encourage them to re-breathe the C02 they expelled, and bring their epinephrine and norepinephrine back to baseline. When you do cyclic hyperventilation, you won't have a panic attack as long as you pay attention to your autonomic state as you do.

If you feel alert after ten breaths, stop. If not, do ten more breaths. If you want, complete twenty breaths, then take a long exhale and hold your breath for ten to Sixty seconds until you feel the gasp reflex, and then start another round of cyclic hyperventilation. When you're as alert as you want to be, resume normal nasal diaphragmatic breathing. Sure, cyclic hyperventilation looks a little wacky but so what?

It costs nothing, and it works. It may also have benefits for your immune system. Just don't ever do it in or near water, or while driving, because when your C02 is low, the impulse to breathe is delayed. People have drowned practicing this breathing pattern as a way to extend how long they can hold their breath underwater. Behavioral Tool 8: Deliberate Cold Exposure Deliberate cold exposure works very well to increase alertness and focus for hours afterward.

That's not surprising: As you now know, it can more than double or triple your circulating levels of epinephrine and norepinephrine and dopamine—our now-familiar friends, the catecholamines—and it does it within seconds. The data on deliberate cold exposure and learning and memory are mixed because most studies have explored if people can learn during deliberate cold exposure, while they are in the cold.34 You can already guess what happens there: Deliberate cold exposure generally puts people at the far end of the inverted U-shape curve of autonomic arousal, where focusing ability sharply drops off. That is not the idea here. This is: Try thirty seconds to three minutes of deliberate cold exposure, then shower and start your learning bout or work session. It's about using deliberate cold exposure to create a long-lasting window of moderate catecholamine increase.

Depending on the duration of the cold exposure, the increase in alertness and focus can last for many hours afterward. The challenge is there is no one-size-fitsall protocol for this one because people's cold tolerance varies. My suggestion is that if you decide to use deliberate cold exposure as a tool to increase focus, select your method, then choose an uncomfortably cold (for you) but safe temperature. Refer to Stress Control Protocol 2 for more information. If you're using a shower, get all the way under the water.

If you're submerging yourself in a plunge or pool, submerge your feet, legs, and torso completely, up to your neck. And yes, put your hands in the water, too, unless you need to grip the sides for safety. The water should be cold, but shouldn't cause pain. Remember that the first ten to thirty seconds are going to feel like mild panic. Your breath Will quicken, but then try long, slow, and controlled inhales and exhales.

three weeks—which is the duration needed to get through the withdrawal—to participate in research. So we are left with very limited studies, all of which point to caffeine as a good way to enhance alertness, but not focus per se. Let me explain. As we discussed in chapter 1, caffeine staves off fatigue by thwarting the action of adenosine, a molecule that builds up in our brain while we are awake and that causes us to feel sleepy

as a result. If you drink caffeine regularly, you're getting some modest focusenhancing effects, as alertness is a prerequisite for focus.

Caffeine also increases adrenaline and norepinephrine in a dose-dependent manner. A single cup of strong coffee—about 250 milligrams of caffeine—can increase circulating adrenaline by more than 200 percent over baseline levels, and norepinephrine 75 percent over baseline 8.6 levels. Ingesting caffeine daily also may increase the density of brain dopamine receptors, allowing the same amount of dopamine to impart a greater effect.8-Z The largest alertness and focusenhancing effects of caffeine are likely to be felt if you don't regularly drink caffeine, or if you were to ingest more caffeine than is typical for you. A note of caution: Remember the inverted Ushape curve that relates arousal and focus? If you are not a regular caffeine user and you suddenly consume 100 to 250 milligrams more caffeine than is typical for you, it Will excessively elevate your sympathetic tone, and you may find it challenging to focus.

I drink caffeine every day, usually in the form of cold brewed or hot yerba maté. I have done that for more than three decades, and by now my caffeine tolerance is very high. Like most people, I swear that it helps me focus though it probably just does that by bringing my levels of alertness up to baseline. If I were to stop drinking caffeine, and push through the withdrawal, I could find my non-caffeinated baseline level of alertness. but I'm uninterested in doing that experiment. I love yerba maté and I love coffee.

I sip caffeine from about an hour after I wake up until about 2:00 pm. daily, and most days I do at least one, and often two or three, long learning bouts or work sessions in that time frame. You'll need to experiment to see what is best for you. I know people who avoid caffeine entirely, those that only use it as a tool for alertness and focus, and I know many like me who use it every day. (Return to Sleep Protocol 11 for more about how caffeine affects your sleep as you experiment with this tool.) Pharmacological Tool 2: Theanine I suggest at some point you try taking 100 milligrams of theanine alongside your caffeine. The caffeine-theanine combination can heighten focus, even for those who consume caffeine regularly.$-$ What's more, the combination of theanine and caffeine has been shown to improve task-switching capacity and reduce susceptibility to distractions," in part because it's thought to increase alpha wave brain activity, which allows for selective attention.90 It also increases levels of the inhibitory transmitter GABA, serotonin, and dopamine, which can help make the alert state created by caffeine feel a little more relaxed. For non-regular caffeine users, it reduces any jitteriness caffeine might cause, as it prevents some

of caffeine's vasoconstrictive and blood pressure elevating effects.91 In fact, even though at doses of 100 to 400 milligrams (depending on someone's body weight and tolerance) caffeine is considered safe for adults without anxiety, taking 100 to 200 milligrams theanine with it may further improve caffeine's safety profile for nonregular and regular users alike.

Theanine may even allow you to get more of a focus-enhancing effect from less caffeine. In one study, people who ingested as little as 40 milligrams of caffeine combined with 100 milligrams theanine experienced improved focus.92 Keep in mind that most commercially sold 12 ounce coffees deliver about 200 to 300 milligrams of caffeine depending on the roast, and 16 ounce coffees can contain as much as 400 to 500 milligrams. Together, the caffeine-theanine combination tends to induce the "alert but calm" state of mind favorable for learning bouts. With its focusenhancing effect, and various health benefits— such as reduced blood pressure and anxiety—l consider theanine a good tool for caffeine users, which means for nearly everyone. Pharmacological Tool 2: Alpha-GPC I consider alpha-GPC a valuable tool for increasing focus and energy levels, if it's used correctly and very occasionally.

Even small increases in acetylcholine—a neuromodulator—can improve focus.23 AlphaGPC does that by increasing the amount of choline that reaches the brain, which in turn indirectly raises acetylcholine levels.24 Research has found an enhancement of concentration and cognitive flexibility in subjects who take alpha-GPC about thirty minutes prior to a learning bout,25 and at least one study showed modest improvement in cognition in people with dementia.96 Other studies, however, show a minimal effect.2Z The variation in outcomes may reflect people's baseline levels of acetylcholine before they take alpha-GPC, similar to what we discussed earlier with people's baseline levels of dopamine before they listen to 40 Hz binaural beats. If a person's natural levels of a given neuromodulator are already high, adding more may not help. If natural levels of acetylcholine are low to moderate, however, boosting it seems to have a positive effect.98 Alpha-GPC is often added to energy drinks, and most nootropic formulas (that is, substances marketed for improving cognition). If you're looking to experiment with ways to increase focus for late afternoon or nighttime work or workouts, and you don't want to disrupt your sleep, you can try 300 to 600 milligrams of alphaGPC taken thirty minutes or so before you begin. Limit your intake to two to four times per week, because your system Will quickly adjust to it.

If you take alpha-GPC, make sure to take 600 milligrams of garlic extract (in capsule form, to avoid the taste and smell of raw garlic) at some point in

the same twenty-four hours, to help offset the increase in TMAO that alpha-GPC can cause.22 TMAO is a metabolite formed by gut bacteria in response to high levels of choline and it can pose a cardiovascular risk if elevated too frequently. Garlic contains an antimicrobial chemical called allicin, which may help offset L-tyrosine is abundant in nuts and seeds, meats, and some cheeses. Parmesan Reggiano has so much tyrosine that you can often see the Ltyrosine crystals on the surface of the cheese block. Yes, it's safe to eat those L-tyrosine crystals, and yes, that is one of the reasons some people like certain cheeses so much—the taste combined with the dopamine increase it causes is highly reinforcing. L-tyrosine gets converted into L-dopa by the rate-limiting enzyme tyrosine hydroxylase, which means flooding your system with Ltyrosine is not suficient to increase dopamine —the amount of tyrosine hydroxylase is the bottleneck on dopamine production.1 2 Nonetheless, there is some evidence that supplementing with L-tyrosine can improve alertness, focus, and learning under certain conditionslO-3 The problem is that most studies used ridiculously high dosages of L-tyrosine to observe these effects—as much as 8 grams or more delivered in capsule or pure powder form.

I do not suggest taking that much all at once! Anecdotally, many people find that supplementing L-tyrosine at doses of just 500 to 2,000 milligrams can help improve alertness and focus, but the published studies indicate L-tyrosine supplements really only help when people are stressed or fatigued.J-Q4 I have tried 500 to 1,000 milligrams L-tyrosine and I've tried Mucuna pruriens, an over-thecounter naturally occurring form of L-dopa. Both increased my alertness, focus, and energy levels for a few hours but, at least for me, they both led to a significant dip in energy afterward. Some people love L-tyrosine at 500 to 1,000 milligrams before work or exercise sessions, and swear by the focus-enhancing effects. That is Why L-tyrosine is found in a lot of energy drinks and pre-workout formulas.

I take it from time to time because it happens to be in one of the energy drinks I sometimes consume. So now to energy drinks Pharmacological Tool 4: Energy Drinks More than a third of the world's adult population consumes at least one energy drink per month, and about 9 percent consumes them every day. Studies of adolescents aged thirteen to nineteen have reported that a quarter of those who regularly consume energy drinks have at least one a day, and another quarter have at least two per day.J-Qé In 2024, the Centers for Disease Control and Prevention noted that 30 to 50 percent of adolescents report some amount of energy drink consumption.JOZ These rates are enormous. To be clear, if you swallow a capsule of 300 to 600 milligrams of alpha-GPC, another capsule of 100 to

200 milligrams theanine, and down both with 100 to 300 milligrams of caffeine from coffee, tea, or diet soda, you're basically making your own energy drink. But there is an important difference.

Many commercially sold energy drinks contain anywhere from 200 to 400 milligrams caffeine, 400 to 900 milligrams alpha-GPC, and 100 to 200 milligrams theanine per serving, and contain one to two servings per can. Many also include L-tyrosine, and more are starting to include huperzine A, an alkaloid that blocks the enzyme that breaks down acetylcholine, leading to overall increases in acetylcholine levels in the brainl Many energy drinks are also now starting to include racetams, a family of synthetic compounds that lead to increased activity of acetylcholine and other neuromodulators in the brain. 1 Now you know Why energy drinks are so habit forming: They are carefully engineered neuromodulator-increasing cocktails that taste like candy. They can even cause withdrawal effects if you regularly drink one or more per day, and then try to suddenly stop. I rarely drink commercially sold energy drinks, preferring instead to get my caffeine from yerba maté or black coffee, and sometimes also taking theanine and/or alpha-GPC in capsule form. If I do have an energy drink, it's usually before a particularly hard workout, as opposed to a learning bout.

I suggest that kids under the age of eighteen avoid energy drinks or pre-workout formulas, which are similar. We have no idea how all that neuromodulator augmentation impacts brain wiring. I hear from people young and Old saying they "can't focus," only to learn that they drink one to three energy drinks per day, or a 20 ounce coffee every morning. That's likely 600 milligrams of caffeine just to feel baseline levels of alertness. Homeostasis is a nonnegotiable rule of biology.

If you want to boost focus and energy levels from caffeine or other substances, a daily high dose is not recommended. Put differently: Whatever stimulant you take every day Will soon be needed to merely reach your baseline levels of alertness. Better in my opinion to use stimulants as an augmentation, not a requirement. That said, I am not going to give up my daily caffeine. I enjoy the taste and the feeling far too much.

I drink just enough of it to bring about ever so slightly toward increased sympathetic tone. It's the perfect state for focus. It also didn't hurt tobacco sales that smoking alongside a stranger or asking for a light was an icebreaker, and that nicotine reduces appetite. Pair all that with advertising that romanticized the act of smoking, and the cost, foul odor, stained teeth, and stagnant breath that smoking causes all went largely

overlooked. Of course, until people finally realized that smoking, dipping, chewing, or snuffng tobacco is dangerous.

It significantly increases cancer and stroke risk.J-!5 And yet, that went largely overlooked for a long time too. Public health announcements started to make these risks Clear in the mid-1960s, but it was a rather Slow and reluctant campaign until the 2000s, when the tobacco industry started to be heavily taxed and regulatedns By 2010, the tobacco industry—formerly one of the most powerful in the world —saw its sales plummet. I was born in 1975 and I can Still remember people smoking in airports and on planes, and how every dining table had an ashtray on it. Even the squat racks in the first commercial gym I joined back in 1993 had ashtrays molded into them so you could set your cigarette down while you performed your set! How times have changed.

Or have they? From 2005 to 2015, as smoking was progressively banned from most every indoor and even some outdoor public spaces, the use of electronic cigarettes, aka vaping, exploded. Vaping is somewhat safer than smoking, but carries its own risks.117 Nicotine itself doesn't cause cancer, though the contaminants in vape solutions cause lung damage. Also, since nicotine is highly addictive, people struggle to quit vaping just as much as they struggle to quit cigarettes. Health warnings about vaping have curbed the market significantly in recent years, but because no other substance creates that alert-but-calm state like nicotine does, as they always have, people make sure to find a way to get their fix.

Which is Why since 2020 the use of oral nicotine products like pouches, gums, and mints has exploded in popularityu-g Notice the pattern? I get asked about oral nicotine almost daily. "Is it safe? Is it addictive? Is using four 6 milligram pouches worse than vaping?" Some people even use it when exercising; millions more use it to work.

An increasing number of women and teens are using oral nicotine products now too. So, here is what you need to know about oral nicotine: It absolutely works to sharply increase focus... if you take it occasionally. Meaning no more than two to three times a week, at a dose of 1 to 6 milligrams each time. If you take nicotine daily, you'll soon need that same dose just to reach your former baseline levels of alertness, and without it you'll be in a mild state of withdrawal. You'll feel foggy and more tired than usual.

And it is very habit forming. That is Why so few people can limit their oral nicotine use to 3 to 6 milligrams once per day. I hear from countless people who started with that amount, loved the effect, then increased to 3

milligrams twice per day, and now use 4 to 6 milligrams or even 8 milligram pouches pretty much all day long. There's more to discuss as you consider this tool. Nicotine may also protect against Parkinson's and Alzheimer's.

You read that right. Nicotine may prevent the neuronal degeneration underlying Parkinson's and Alzheimer's disease.U-2 Clinical trials show, however, that once these diseases start, nicotine is unlikely to be of benefit.U?-Q Whether any neuroprotection that nicotine provides is great enough to offset the other health problems nicotine is known to cause is also Still unclear. Remember, nicotine itself doesn't cause cancer— it's not mutagenic. But it may accelerate tumor progression.J2-J It also raises blood pressure and causes vasoconstriction, which increases heart attack and stroke risk, and because of its vasoconstrictive effects, it has the potential to cause sexual side effects too, especially in Experts on oral health are noticing that patients who use nicotine have more gum disease. Experts on skin health observe that it may cause premature aging of skin.123 Both are likely as a result of its vasoconstriction effect (ie., reduced blood flow to the gums and skin).

That all makes sense, as reducing the blood supply means less oxygen and nutrients get delivered to an organ or tissue, and fewer waste products of cellular metabolism are removed from it. Curious about the oral nicotine hype, I tried 4 milligrams nicotine gum and it worked to make me feel alert but calm. And it worked quickly, like flipping a switch. The next day, however, I felt a bit less mentally Sharp. I also felt I needed to Clear my throat a lot.

I spoke to an MD friend who works on neural pathways for speech, and he told me that nicotine binds to both nicotinic and muscarinic acetylcholine receptors—the latter of which are abundant in the smooth muscles of the throat, larynx, and pharynx. He explained that after nicotine wears off, those muscles go into a slight spasm. Sure enough, taking 4 milligrams nicotine again fixed all that right away... until the next day. That experience was enough for me to end the experiment. Fortunately, the throat issue resolved after about three days.

I share this story because it made me understand the appeal—the focus-enhancing effect of nicotine is remarkable—and yet, people should know all the effects, which 1'11 recap one more time here to underscore them. There's addiction potential, increase in blood pressure and vasoconstriction, and of course how quickly the cravings begin and many of the positive effects evaporate. I have friends who are otherwise very health

conscious, and they need nicotine to do any sort of focused work or exercise. It's become part of the way they just get to baseline. Are they addicted?

The classic definition of addiction is continued use despite harm to self or others. One way that I define addiction is something that causes a progressive narrowing of the things that bring you pleasure. According to both definitions, they are not addicted to nicotine (unless you consider the long-term health harms). But I do think they are dependent on it. As with all things, my motto is this: Do what you want, but know what you're doing.

If you want to use nicotine, use the lowest possible dose that is effective for you. Stick to pouches, mints, or gums because the cancer risk appears to be lower than with cigarettes, but be aware that we do not yet have enough data to know this risk conclusively.12 Limit your use to two to four times per week, and at most 6 milligrams per use (2—4 milligrams would be even better). That is, know I am not an MD, but they tell me it is hard to find a US doctor who doesn't almost immediately default to prescribing these medications. They are confused and conflicted about what to do. Here is what everyone should know: All prescription drugs for focus have side effects.

These vary by drug, dosage, and by person in ways that can only be determined empirically, meaning by trial and error. Also—and this is very important—the dosage at which some of these drugs are effective is enormously variable. Adderall works to treat ADHD in some people at 5 milligrams, whereas others need 40 milligrams It's simply impossible to know or even more. how someone Will respond, and it requires time to find the minimum effective dose. Of course, all the drugs in this section require a prescription to acquire legally. Illegal and questionably legal sources carry significant risk.

Fentanyl—a deadly opioid—has frequently been found in counterfeit Adderall.J2-Z The reason I call these "the heavy guns" is they work so well that people quickly become dependent on them, and some, such as the amphetamine-based stimulants like Adderall, carry abuse potential because they can create It's also important to note, however, that failure to treat kids or adults who have ADHD puts them at a much higher risk for addiction than not, due to the impulsivity associated with ADHD.U?-2 That said, amphetamines may stunt growth,L30 cause sleep disturbances,131 and even lead to psychosis if abused or taken by people with bipolar depression or a predisposition to psychosis.J-3-Z And all these drugs are hard to quit.J-3.3 When someone does try, they often suffer from excessive

daytime sleepiness and mood and focus issues, aka withdrawal. It's much worse than caffeine or nicotine withdrawal, which is Why most people don't quit—they use these drugs indefinitely. Thus, the decision to medicate a Child or adult with Adderall or Vyvanse (both amphetamines) or Ritalin (a non-amphetamine stimulant) needs to be carefully considered. There are also a lot of nondrug therapies for ADHD that can be effective and/or can help reduce dosage needs. They are the foundational and behavioral tools mentioned above.

What about supplements? Can alpha-GPC replace a genuine need for Adderall to treat ADHD? Unlikely, but a lifestyle that follows the foundational behaviors described in the protocols here (sleep, hydration, nutrition that favors stable blood sugar levels and satiety, etc.) along with regular meditation or hypnosis for increased focus, and timed use of caffeine and theanine, may allow people to use a lower dose of a prescription drug for ADHD. There are also many cases where these lifestyle changes have allowed people with ADHD to stop taking their medications and Still thrive, or thrive even more. It's all highly individual.

That's ADHD. What about the person who doesn't have ADHD, but wants to use a prescription medication to induce states of intense focus one to three times per week? They may be doing everything else correctly, adhering to the foundational tools I've described in this chapter, but they want an "edge." And what about the many millions of people who, due to smartphones, have what clinicians are now calling mild ADHD, which is not a diagnosis but a catch-all for no longer being able to focus?

Should they take a sympathomimetic? We can no longer pretend that smartphone-induced mild ADHD doesn't exist. My stance is that we should be aware of all the tools available for focus that are legal, and consider their relative potency and risks. RITALIN This non-amphetamine stimulant has largely fallen out of favor for the treatment of ADHD and for focus purposes because it's short acting, and people prefer the stronger, longer lasting effects they get with Adderall and Vyvansel A recent brain imaging study found that methylphenidate —the active ingredient in Ritalin—did not actually increase activation of attentional networks in the brainl It merely reduced the feeling of fatigue and the effects of sleep deprivation on attention. It also made boring work feel more tolerable, and even rewarding— thus allowing for more sustained effort at tasks.

The study didn't look at other stimulants, but this may be a general theme of the drugs covered in this section because few if any stimulants directly increase focus. Raising dopamine levels can reduce background "noise" in

the brain, and therefore the tendency to get distracted, and raising norepinephrine increases alertness. Together, that can make trivial things feel more engaging. It's important to know that people with ADHD can focus quite well—they just have a much harder time focusing on things they don't enjoy than do people without ADHD. Alas, much of life requires focusing on things we may prefer not doing.

Apparently, stimulants like Ritalin trick the brain into ignoring our preferences.

amphetamines. (Also, see the section "A Note on Phone Management" at the end ofthis protocol, as the Phone is a severe impediment to focusing for many people.) To be clear, I am not against prescription drugs per se, and if you do take one, don't stop suddenly. Talk to your doctor. Many prescription drugs have their place in treatment, but unless there is a genuine clinical need, using stimulants as powerful as Adderall or Vyvanse on a regular basis Will soon cause dependence, and other tools you try Will pale in comparison. MODAFINIL AND ARMODAFINIL These drugs were designed to treat "excessive daytime sleepiness" caused by narcolepsy or sleep apnea or jet lag. Today, they are commonly prescribed to people simply wishing to improve their focus for work or school.

They are not amphetamines, but they can similarly disrupt sleep and appetite, and they Still pose some cardiac risk—a feature of all sympathomimetics because they elevate blood pressure and heart ratel These drugs are interesting because while they raise alertness and focus, they also have the rare feature of also improving cognition, in particular executive functions that allow for orchestrated task switching and working memory—the ability to hold relevant information in mind for only as long as is necessarym At the same time, they may impair creative divergent thinking.U4.2 In speaking with psychiatrists who prescribe these drugs, I've learned that people's sensitivity to modafinil and armodafinil varies tremendously. Some find 50 milligrams much too stimulating and their sleep is disrupted; others need 200 milligrams to achieve the benefits and sleep just fine. I have tried both armodafinil and modafinil and found they can erase my jet lag but they impair my recall memory. Armodafinil is essentially longer lasting modafinil, but even at the lowest dose of modafinil (I broke a 100 mg tablet in half), I struggled to fall asleep later that night. That said, I have a colleague who takes 100 to 200 milligrams of modafinil four to five times a week, which was prescribed to him for "mild ADHD," and he has no issues sleeping afterward.

These drugs used to be expensive—but the release of generic versions made it less than one dollar per 100 milligram pill. That, combined with their mild cognitive-enhancing effects, makes me think that modafinil and armodafinil use Will increase in years to come. SUNOSI This is a relatively new drug that also acts to increase dopamine and norepinephrine in the brainl It is FDA approved for offsetting excessive daytime sleepiness due to sleep apnea, and recently it performed well in clinical trials for ADHD,I making it likely to soon be approved for that purpose as well. I know people have already started using Sunosi by off-label legal prescription either daily for ADHDJ-4.5 or occasionally for enhancing focus. Sunosi is not an amphetamine, and thus has less abuse or addictive potential than the other compounds mentioned above.

It does not cause euphoria, nor does it seem to cause withdrawal on days it is not taken. Provided the dosing is correct (typically 75—150 mg taken in the morning), it does not seem to disrupt sleep either. Interestingly, Sunosi is a much lower amnity dopamine and epinephrine reuptake inhibitor than Wellbutrin, Adderall, or modafinil and it may have some mild serotonin-increasing effects too. That might explain Why people find it less stimulatory and anxiety promoting than other pro-alertness and focus drugs. As with all drugs that increase dopamine, people with conditions that cause mania or psychosis should be cautious about using Sunosi.

WELLBUTRIN This dopamine and norepinephrine reuptake inhibitor is much more noradrenergic than dopaminergic as compared to the other compounds in this sectionl This means it raises norepinephrine more than it raises dopamine. It was developed to help with smoking cessation, and to treat depression, while bypassing the sexual side effects caused by many SSRIs. Eventually it was adopted by the ADHD community—a mix of psychiatrists and online patient groups that found it useful for its offlabel use to increase focus.J4-Z At higher dosages —200 milligrams or more—Wellbutrin increases risk of seizures in those who are seizure prone, and it carries the usual anxiety and cardiac health warnings of all sympathomimeticsu-o Because of its low dopamine-promoting activity, it has low abuse risk. I know many people who take 100 to 150 milligrams of Wellbutrin daily—typically the sustained release versions—purely for its energy and focus increasing effects. I have taken 75 to 150 milligrams specifically for that purpose and it worked well, but I found that Wellbutrin focus on the task at hand, having their Phone within physical reach reduced their cognitive ability.

Apparently, even if we are not consciously thinking about our Phone or seeing notifications, if it's within our physical proximity our brain devotes

mental energy to resist the temptation to pick it up. The takeaway thus is simple: To maximize focus, put your smartphone in an entirely different room from where you are. Another very useful tool is to restrict your social media apps to an Old Phone that has a completely different Phone number. I did this, and now I purposefully limit my time with my "social media phone" to sixty minutes or less per day. If you do both those things and Still find yourself continually looking at your phone, I suggest getting a timed lockbox that has a no-code-out option—meaning there is no way whatsoever to open the box until the timer has elapsed.

I use a no-code-out lockbox for my phone, and find it helps for getting work done. I don't have ADHD but I find that once the Phone is truly inaccessible, my ability to focus is massively increased. The best analogy I can come up with for the smartphone as it relates to the topic is this: Just as sleep apnea or daytime apnea reduces the amount of oxygen reaching your brain, smartphones cause attentional and cognitive apnea. They lessen the amount of neural activity in our brain circuits for focus. To focus on a work session or a learning bout of any kind, get your Phone far out of reach.

Your mild-to-severe ADHD may be more phoneproximity-driven than you realize. Perhaps partially, perhaps entirely. There is only one way to find out!

LEARNING PROTOCOL Use Mental Contrasting to Achieve a "State of Urgency" and Reach Your Goals

WHAT TO DO

• Think of being in "a state of urgency," rather than a state of motivation. Urgency is an experience we are all familiar with. We tend to know what it feels like in our mind and body. Motivation can be vague and ephemeral. • The most reliable method to create this state is a psychological tool called mental contrasting. It asks you to imagine your goals, and the obstacles to achieving them, in a structured way.

Even if you're already on your way toward your goals, this strategy can further assist you.

HOW IT WORKS

"Motivation is everything." We hear this all the time and know it's true, yet people vary in their abilities to get and stay motivated. Especially to do things that require sustained focus and energy.

purpose. Alia Crum, a professor of psychology and researcher who directs her own lab at Stanford, explained it best: "A mindset is a setting for your brain. Meaning, a mindset is a psychological stance that makes certain thoughts and actions easier to access, and other thoughts and actions harder to access. Alia's definition also speaks to the power of mindsets to simultaneously influence different aspects of your physiology—your brain directs many systems, including your nervous system, your hormones, even your immune system, in specific ways.J-52 That, in turn, guides our behavior. When we switch to a mindset that embraces a challenge, it activates neural circuitries and chemical reactions that make us feel more capable of running that extra mile, putting in extra study hours, or going for one more hour of piano practice.

Physiology is also a lever here, but a secondary one. You may have heard the terrn "growth mindset," coined by Stanford psychologist Carol Dweck. The term refers to the belief that ability is not fixed but rather can improve with effort. 1 With a growth mindset, a teenager who is failing calculus understands that studying hard Will lead to better comprehension. That state unlocks the possibility for even more: You believe in the effort that learning requires, and take the action that it requires too. Upholding a growth mindset means you actively avoid identity-based and valuation labels of being good or bad at something.

You aren't "good" or "bad"; you simply haven't put in the effort and taken the action yet. A growth mindset never says, "I'm not a math person"; it says, "I can get better at math if I work at it."

Experiments from Carol and others show that adopting a growth mindset can drive significant improvements in learning, especially when paired with a stress-can-enhance-performance mindset.L54 That last bit—that stress can help with growth—implies that challenge can improve your outcome, rather than limit you. Maybe you are wondering: If stress is so great for learning, then Why aren't we all great learners? We all certainly feel stress. I think it's because for most of us, the feeling we have when we fail at doing something dificult overrides the desire to keep striving. That is one reason I continually share ways for real-time autonomic selfregulation, like the physiological sigh described in chapter 3.

These protocols are designed to lower stress not only to help people feel calm, but also to expand people's sense of what's possible to learn. Ifwe have the tools to control stress, we can seek out more challenge. No matter how well we control our stress, however, we are Still left with the challenge of how to activate a growth mindset and

stress-canenhance-performance mindset. In other words: How do we actually get motivated? Should we use self-talk?

Watch an inspirational clip on YouTube? Fantasize about achieving our goal? Scare ourselves with the fear of not achieving it? How many milligrams of fear should we take? Just kidding.

Sort of. The answer starts, I believe, with swapping the word motivation with a state of urgency. Here is why: • A state of urgency is less prone to parsing than motivation, which immediately raises questions about context. Friedrich Nietzsche offered powerful wisdom when he wrote, "Hat man sein warum? des Lebens, so vertrãgt man sichfast mitjedem Some readers may be more familiar with the English translation made famous by Austrian neurologist, psychiatrist, and Holocaust survivor Viktor Frankl: "He who has a Why to live for can bear with almost any how."J-5.6 • Sometimes the Why is clear. But what about when it isn't?

What about the many aspects of learning and life that require persistent effort and for which the why, the deeper meaning and larger purpose, is unclear? Often the Why only becomes Clear during, or even after, the how. Maybe a person is forced to take a course they hate in order to get a degree. Once they complete the course, however, they may realize the new opportunities have been made accessible to them only because of the material they learned in the course; it wasn't just a means to an end after all. A state of urgency doesn't require you to have all the answers yet. • A state of urgency stimulates catecholamine release, which further increases sympathetic tone, which therefore increases focus.157 This helps dim distractions, both internal and external, which improves performance... t'sa and increases your state of urgency.

I positive feedback loop. • It's obvious Why a sense of urgency would help you maximize the number of repetitions that you can do in a given time frame. That's huge because genuine attempts are the fundamental unit of all forms of learning.

All that is consistent with what we know about dopamine in the context of goal seeking and rewards. As Anna Lembke, an expert on dopamine, put it, dopamine is about wanting, not about having.L6.3 That's another downside of positive thinking: It activates the "want" circuitries but not the "have" ones. Anna further pointed out that the neural circuits underlying motivation evolved in times of scarcity, where effort almost always preceded reward—think hunting, gathering, seeking shelter, finding a mate, and so on. Because of the abundance of stimulation in modern life, which was designed to satisfy and even exploit our dopamine circuitry,

most of us have recalibrated our motivation and reward circuits to expect more reward from less effortl And while Anna's work has mainly focused on addictive and compulsive behaviors, we've all seen this play out for goals tool Too much reward prior to effort, or too much reward without effort, undermines future effort. A friend of mine, an MD turned successful startup investor, warned me to be careful about telling too many people about a big goal.

Most Will respond, he explained, by saying "You would be great at that, go for it!" Unless you have a lot at stake, the reward of all that positive feedback can undermine your drive to actually pursue the goal. I asked him if he thought it would be better if we responded instead by saying, "Don't try, you'll never succeed!" "I don't think that helps either," he replied. He was right. Oettingen found that if people focus too much on the challenge they'll likely face, that too prevents them from pursuing goals, at least in the long run. So much for just "Focus on the journey not the destination." We can thus all be grateful to Oettingen for carrying out experiments that tested various common motivational tools, alone and in combination.

She also explored their emcacy for helping people achieve different types of goals: learning cognitive skills, learning physical skills, making better food choices, exercising more, and on and on.U6.6 She eventually discovered that it is the contrast between visualizing the goal and thinking about the obstacles to achieve it that creates a state of heightened physical and mental energy, and that state is what leads us to be more likely to start—and to persist. 1 In other words, Oettingen figured out how people can activate a state of urgency. She calls the approach mental contrasting, and she calls the way to implement it WOOP, an acronym that stands for Wish, Outcome, Obstacles, Plan.J-6-g The four steps of WOOP can be done in your head, but it works best if you record your answers. Try writing them down, or hitting the record button on your phone. It works for both short- and longterm goals. WISH: Name the big, overall goal.

What do you want to happen? For example: • "I want to learn guitar well enough to play with a band live in front of a thousand people."

• "I want to get a PhD in neuroscience from a top program." • "I want to learn to speak fluent German." • "I want to paint well enough to make a living selling my art." It takes only a few seconds to complete the W step in WOOP. You don't need to include a lot of detail, but some helps. For instance, "I want to be happy and successful" is too general. Instead of focusing on what you want to feel, name skills, credentials, and proficiencies that you're hoping to achieve. OUTCOME: Here is where you

think about what you want to feel from having achieved the goal.

For example: • "I Will experience the thrill of performing live with other great musicians and knowing that I'm giving people the live music experience I know and love so much too." • "I Will feel the freedom and excitement ofbeing able to work for any number of different companies exploring for cures to neurodegenerative diseases." • "I Will enjoy going to Germany to visit my spouse's family, be able to relate to them better, and maybe even work and start a life there someday. And I can help our kids become bilingual." • "I Will finally get to indulge my desire to be creative. And I can sleep in late and stay up late painting while listening to music, which is what I always wanted!" You'll notice this outcome step is aspirational— you're encouraged to daydream about how you want to feel. I gave short examples here but feel free to expand on the many things you look forward to feeling. The reality check is next.

OBSTACLES: Get Clear on the obstacles you anticipate en route to your goal. Be hyperrealistic. Record the roadblocks you might face. Include how you might feel as you do. People's temperaments vary, but you know yourself—so be honest about how you'll react when things don't go to plan.

For example: • "I tend to be really tired from work at the end of the day. I'm going to feel frustrated when I am too tired to practice guitar or when trafic makes me late to guitar lessons." • "I Will miss the chance to attend weddings and go on fun vacations with my friends because 1'11 be in the lab doing experiments or studying. I won't have much money because PhD students don't get paid much." • "It Will be a long time before I am truly fluent in German because I learn really slowly. And writing in German is even more dificult than speaking it, so I Will get discouraged a lot." • "The art I make is not likely to appeal to a lot of people because I like melancholy themes and darker tones, and most people seem to buy art that is happy and playful." Again, feel free to expand as much as you want to for this step. PLAN: Come up with if-then statements to address the obstacles you anticipated.

These are action steps. For example: • "On days when I'm too tired to practice guitar after work, I Will go to sleep early and get up thirty minutes earlier than usual the next day. That way I can get fifteen to twenty minutes of practice done in the morning. My mind is much sharper then anyway. Playing in the morning also makes me feel good for hours, so maybe work might not feel so exhausting.

If I'm late to guitar practice because of tramc, then I Will do my best to stay calm and practice in my head on the way there. I Will add a few

minutes of practice in my car after the lesson to really drill in what I learned." • "When I can't attend fun events because of time or money restrictions, I Will remember that having a PhD in neuroscience puts me at a unique advantage for job choices and longterm financial security because there is so much Still to discover about nervous system diseases... and we Still don't have cures to most all of them. Also, a lot ofmy friends who make more money than me now are bored with their jobs or have already maximized their earning potential. I Will be able to consult and/or invest in biotech companies because I really understand where the field is and where it might grow. Also, I Will remember that I love doing experiments." • "If I feel frustrated at my Slow progress learning German, I Will remind myself that the frustration is a signal my brain is getting the message to change, and that a lot of learning happens in bursts; it's rarely linear.

I Will ask my spouse (who is fluent) ifwe can do an only-in-German dinner once per week I followed the WOOP process. I typed out the WOOP answers into a word processing document and hit Save. That took about ten minutes. And it worked spectacularly well for me. It somehow cleared away the laziness and analysis paralysis and all the other stuff that had been holding me back, and I enjoyed two solid work sessions that very day, and then two months of steady effort toward the goal thereafter, even though I was in my busiest season of work and I had many other commitments.

When I started feeling a bit unmotivated again, I reread the document. I made some edits and additions based on new changes in life. That only took about five minutes. I now do mental contrasting for learning goals, for work goals, for fitness goals, and for personal goals too. I reread and update the WOOP documents whenever I am feeling a bit unmotivated.

It's become reflexive. I don't share my WOOP answers with anyone, they are for me and for me only, but you can decide if you want to keep yours private or not. The Mechanisms Behind Mental Contrasting I was struck that mental contrasting seemed to increase my motivation across the board, not just for my specific goal. I procrastinate far less on everything now. I'm not aware of any data showing that doing mental contrasting rewires specific brain areas, but given what we know about the aMCC, I wonder if it might.

Recall how Joe Parvizi conducted incredible experiments showing that if activity in the aMCC is increased, people feel an immediate surge in mental energy and readiness to lean into challenges. Especially since every instance of learning—whether that's a new behavior or a new piece of

information— is the consequence of plasticity and our aMCC is the brain structure most closely associated with tenacityl It has been shown to increase in volume in successful dieters, people who push themselves to Stick to a new exercise program, and other situations where thinking about and overcoming a challenge is called for. Parvizi discovered the role of the aMCC in 2013, and only recently has it been discussed in the larger field of neuroscience and media, so I doubt that Oettingen was aware of its potential role in WOOP. I look forward to a day soon when someone explores whether the aMCC is activated by going through the steps of the WOOP Protocol. In the meantime, I plan to keep using mental contrasting any time I need to generate a state of urgency for something, and I hope you'll consider it too.

Why Some People Will Avoid Mental Contrasting This may surprise the dreamers and go-getters, but a lot of people are afraid to strategize around —let alone think about—their big goals. Some of that is culturaP-ZO but in my observation a lot of people are afraid to change, even if they are unhappy with their current circumstances. It's just not reflexive (and in many cases it feels aversive) for them to do so. Before you conclude that a resistance to change is just the way you're wired, consider the experiment where people preferred to get strong electrical shocks they knew were coming at specific times... versus weaker electrical shocks that arrived without warningm The same general conclusion comes from studies of pain. People like what's predictable, even if it isn't good.

That deñes logic until we remember that our nervous systems are prediction machines that seek to minimize energy expenditure.172 so, while mental contrasting is merely a mental exploration of a goal, I anticipate some people Will Shy away from it... because change is hard. I hope you don't and instead, go with the data. Put it to work for you. WOOP is a really cool tool to explore new things, and I'm grateful Gabriele Oettingen and colleagues came up with it. And learning to switch on a state of urgency to increase focus and achieve your goals and get things done in life is a valuable superpower, for SDAP and beyond.

LEARNING PROTOCOL Embrace Errors to Improve Your Performance

WHAT TO DO

• Embrace the fact that errors are an essential trigger for learning. Errors that come from genuine attempts to learn something are the signal for plasticity to occur. This is true for all forms of skill development: cognitive, motor, musical, mathematical, et cetera, and Why it's an essential step in

SDAP. • Strive to perform at a level of dificulty just slightly beyond your current ability. Studies show that the right level is such that it produces an error rate of 15 to 30 percent. The 30 percent level tends to work best for learning motor skills, whereas 15 percent is the approximate level to shoot for with cognitive skills. • Don't make errors deliberately.

That won't have the same result. Work on pushing yourself past what you think you can do and using your mistakes as learning tools.

HOW IT WORKS

If you can do something with proficiency, your nervous system won't change. The gap between shelter, potential mates, et cetera. Unwelcome surprises, even if they are not traumatic, shape our memories, help us learn, and guide our future actions too. Whereas positive events simultaneously increase dopamine and lower serotonin, negative events do the opposite. Say that instead of meeting a new friend at a party, you unexpectedly run into an Old boss you hoped to never see again.

That would shape your expectations and motivations around future gatherings at that house. It may even change your feelings about the friend—a phenomenon known to neuroscientists as context-dependent plasticity.17 Why We Need To Make Mistakes You'll notice that none of the examples of learning I just described require that we make errors. Stuff happened, for better or worse, and learning occurred. It speaks to the many types of plasticity. SDAP is different.

We need to make errors. With errors often comes some amount of agitation, of limbic friction, which by now you know is a good thing because it increases the release of catecholamines that facilitate focus and plasticity. But also, perhaps more importantly, mistakes provide the signal that sets the size and type of neural circuit changes (and thus skill improvement) that follow learning attempts. 1'11 use motor skills as an example but the same holds true for all types of skills. Let's say you want to learn tennis. You need to learn to serve the ball.

At first, it's likely that you'll hit the ball into the net often. Each time you do, it's an error that you register as objective performance feedback. You adjust your serve accordingly. You may decide "that last serve was too hard, 1'11 use less force this time" or "perhaps I am gripping the racket too high on the handle, 1'11 hold it lower" or "I'm throwing the ball too high before I swing," and so on. Even ifyou have a tennis coach who gives you instructions, you are the only one who has access to the proprioceptive

information associated with each serve attempt—by that I mean the awareness of your body's movements.

Your nervous system tracks each and every one of those, and compares it to the performance outcome. Some of that is conscious, and some is unconscious, working below your level awareness. Remember back in chapter 3, when discussed what your nervous system does? you may recall, whether you are asleep of we or awake, your nervous system is handling the basic stuff of survival, making sure that your heart keeps beating, your lungs keep expanding, your digestion keeps flowing, and so on. It is also letting you perceive the world around you. And, as we also discussed, your nervous system is also a prediction machine.

It's constantly trying to make causal inferences: "If I do A, then B seems to happen immediately afterward... If I don't do X, then Y happens a few hours later..." When you are awake, your nervous system is running a ceaseless if-then analysis of you and your actions and feelings, thoughts and memories; of others and their actions and feelings, thoughts and memories; of the world around you, and on and on. Here is the key thing to remember: Any time your nervous system figures out that one of your actions leads to a successful outcome, it starts to shift the pattern of neural firing that occurred from your conscious awareness into your implicit motor memory. It works to make the action automatic. The rationale for that is obvious: Thinking and doing and making associations is metabolically demanding.

As is rewiring brain circuitry. 1 The more your brain can make things automatic, the more energy it can devote to other things. This is especially true in the case of skill learning. With maintaining focus, moving into a state of urgency, making errors, and calibrating adjustments, we can all learn to execute new skills. Once we do, we don't need to think about each step. We simply activate performance of the skill.

We serve the ball, play the chord, finish the math problem. Voila! I acknowledge that your reaction to this point might be: "Duh!" But I said there was a key thing to remember. It Will forever change the way you approach learning. First, your brain doesn't work alone here.

It needs your second brain. No, not your gut. I'm referring to the mini brain tucked right under your occipital lobe. If you touch the top of your neck where it meets the back of your skull, you're touching the area just above your cerebellum, aka your mini brain. Many people have heard of the cerebellum for its role in balance, but your cerebellum is also crucial for eye movements related to attention, for proprioceptive awareness, for

speaking, for estimating time, for your sense of rhythm, and much more.

Your cerebellum is also a key Site of action for comparing what you are trying to do with what you're currently doing, and sending the rest of your nervous system the signals it needs to close the gap in performance. I would love to devote a thousand pages to the function of the cerebellum in learning because it is that cool, but 1'11 keep it (relatively) brief. When you make errors in the context of trying to learn a new skill, you activate neurons in your cerebellum. The bigger the error, the larger the response from your cerebellum. Your cerebellum is kind of like that tennis coach in this regard, except unlike a coach that observes you and makes educated guesses as to what you need to do in order to improve, your cerebellum has full access to both your inner and outer workings: the angles of your limbs and joints, the speed and stretch of your muscles; it knows when you blinked relative to your maximum effort, your level of autonomic arousal, et cetera.

Thus, your cerebellum is also uniquely suited to be your coach because of both the specificity and the interconnectedness of its wiring. Different cerebellar neurons respond to the movement of different parts of the body, even your eyes and lips and tongue. It also encodes how movement of one body area changes the movement of all the others. If you consciously adjust the angle of your elbow and it improves your serve, your cerebellum registers the many other changes that accompanied the elbow adjustment and then makes sure they become part of your serve "motor program" going forward. If your elbow angle adjustment makes things worse, your cerebellum registers that too.

With repeated effort, your brain eventually stamps down the exact motor program needed to serve the ball properly.

the right strategy for neural circuit modification. It was much better, the researchers found, to continually adjust the training level to maintain the approximately 15 percent error rate. This was a quantitative modeling study, meaning the findings are based on a simulation, but it was aligned with studies of visual-perception-based decision-making and motor learning. It works for cognitive tasks too. For example, students who are allowed to figure out how to solve complex math problems on their own—and therefore can make mistakes—tend to do better than students who are told the right way to do it from the startl And we seem to know this intuitively.

In one study, researchers at Columbia University asked fifty-two people to play a series of computer games. I They were free to choose from a range

of games, some of which were easy, some harder, and some impossible to learn. As the task progressed, participants chose increasingly dificult games and preferred to repeat the challenging games rather than repeat the easy ones. Remember the 15 percent "rule" is an approximate measure. There's no need to become obsessed with maintaining this exact error rate.

To put it in simple terms, you want to work at a level at which you're getting things right much of the time, but also wrong some of the time. This insight can be used in education as well. Given the relationship between errors and plasticity, and the pivotal role of the cerebellum in that process, teachers should consider reserving verbal feedback to students in cases where the students thought they got something right but actually didn't. Otherwise, it's better to give students the opportunity to register their own errors. The moment immediately following making a mistake is critical for learning.

Negative reactions typically occur within 100 milliseconds,J-80 but the sympathetic arousal lasts longer and provides the milieu for enhanced focus and plasticity as we have discussed many times now. Which means our systems are primed for learning in those immediate post-error moments. This internal state improves the effectiveness of the whole learning bout. Studies of motor learning show that within-trial errors and across-trial errors both provide valuable signals for plasticity and learningm Ifyou leave a language Class or tennis lesson or other learning bout thinking, "Darn, I couldn't seem to get that right on that one part," well, congratulations! The learning process set in motion by within- and across-trial errors was triggered.

You'll no doubt come back better next time... but only if you do what's in the next protocol.

LEARNING PROTOCOL Prioritize Sleep the Night After a Learning Bout

WHAT TO DO

• Get adequate sleep the night after a learning bout. That's when plasticity actually occurs, and Why it's an essential step in the SDAP process. • Enhance plasticity by practicing ten to thirty minutes of NSDR at any point after a learning bout. Done prior, as described in Protocol 1 of this chapter, it Will increase your mental and physical vigor.

HOW IT WORKS

During sleep, our brains rapidly replay what we tried to learn during the day, including what errors to avoid in the future. Brain imaging and electrophysiological recordings show that the very same neural firing sequences that occurred during learning bouts occur again at that timel-g-a If the patterns of brain activity in sleep are disrupted, we don't learn as well. Sleep isn't just a time for consolidation, it's also when we build schemas—which are extractions of common themes of movement or ideas that let our brain store information in a more eficient way. They serve as patterns that become part of our LEARNING PROTOCOL Incorporate Micro Rests into Your Learning Bouts

WHAT TO DO

• Pause now and then between repetitions as you practice a new motor or cognitive skill. These waking micro rests have been shown to accelerate learning by increasing repetition density at the level of neural circuit activation. • Short breaks of five to thirty seconds during a learning bout in which you do nothing at all (meaning, no checking your phone!) can significantly accelerate motor learning. • Short breaks of five to ten minutes during a learning bout focused on cognitive learning can do the same. These micro rests induce ten to twenty times accelerated replay of the neural circuit firing pattern related to what you are trying to learn.

HOW IT WORKS

We try to perform a new skill and we fail, and fail again, and get close, but fail again. We get a few nights' adequate sleep, and when we come back for our next attempt, somehow we've improved.

the participants' brain waves. During very brief periods of rest between repetitions—lasting about ten seconds—the patterns of brain activity showed a unique signature. They tended to occur in regions connected to our ability to plan our movements. The researchers hypothesized that some form of additional rehearsal was occurring in their brains during the micro-rest periods, and it might be important for plasticity. That mapped well onto Ebbinghaus's observation.

A second study, carried out in 2020 by Ethan Buch and others, found that micro rests induce rapid plasticity and learning, most likely because the same neural circuits active during the attempt to perform the motor skill get replayed during the micro rest in between repetitions, but at about twenty-fold the speed and forward and backward.U8.6 In other words, waking micro rests increase repetition density. This wakeful consolidation

seems to allow you to integrate the new motor sequence with prior learned sequences. Of course, it's during sleep when our brains really engrave the circuits carrying newly learned information. Nonetheless micro rests of ten to thirty seconds during motor learning bouts, and five to ten minutes during cognitive learning bouts, have been shown to improve many different types of skillsl-3-Z You may be wondering what to do during the micro rests, and how often to take them. For a learning bout focused on motor learning, the key is to simply pause and not do another rep for five to ten seconds.

A bit longer, say fifteen or seventeen or twenty seconds is not a problem, so don't obsess about the exact duration. Just pause for a short time, and then continue. You can draw a bit or just sit there—which sounds boring and might make you feel lazy, but the data say you are learning faster. Yes, you can get up and walk around, use the bathroom, et cetera. Just stay off your phone, web browser, or email...!

Introduce micro rests somewhat at random. How often you do them depends on the complexity and the duration of a repetition. Practicing serving a tennis ball is very different, for instance, from practicing an entire song on the piano. I suggest you start the learning bout by doing continuous repetitions one after the Other, and when you start to make errors, introduce a pause every third or fifth trial, then go back to continuous repetitions, and then pause after every error or every other or every third error. If you want, you could set a chime to go off at random intervals.

A website that offers that feature for free is wwwrandomorg/clocktimes and there is another great, no-cost website at wwwmicrobreaksco/timerhtml that lets you set the parameters, including (1) learning bout duration, (2) duration of micro-break, and (3) the tone you want to hear, et cetera. It's terrific for both cognitive and motor skill learning. (Note: I have no afiliation to it but I use it.) Waking micro rests are a great tool for accelerating learning and plasticity that are based in behavioral and neuroscience-backed data. I suggest trying them.

LEARNING PROTOCOL Increase Retention of New Skills and Information with Mental Rehearsal

WHAT TO DO

• Take one to five minutes to reflect on what you read or did during a learning bout at some point after the learning bout concludes. The key is to

test yourself for what you do and do not remember, do or do not comprehend. That's Why it's the final step of SDAP. • For cognitive skills and new information, this can be informal, done simply as a mental exercise, or more formal, such as by having a friend or ChatGPT test you for comprehension on the exact same material. • For motor skills, think about where you were challenged and made errors.

HOW IT WORKS

All of learning is anti-forgetting. On the one hand, that statement is another giant "DUH!" But through the lens of what we work with it more flexibly. That tells us that what we remember is not simply a matter of how much time we spend with information or in an experience, but that reflection somehow reinforces and enhances the circuits carrying that information or experience. This research has vast implications for educational settings, particularly for how best to study. We know that underlining and highlighting information only helps us remember it if we also later write down what was highlighted and underlined.

It also means that testing—whether by our teacher or ourselves —is not just a form of evaluation to gauge our level of knowledge, it also represents a valuable "anti-forgetting" tool that helps us remember the information for months or years to comel I The takeaway is clear: Tests work. If you're a teacher or coach, yes, this justifies pop and scheduled quizzes. It may help to test on recently learned material in exercises that count toward a grade. And, yes, don't tell the students that until after the quiz. For those without a teacher or coach navigating SDAP, reflect on what you learned, and most of all: self test!

You can do so informally or formally. A1 is a great tool for this. Ask ChatGPT to quiz you on a given set of material as a teacher would. Or simply take a few minutes and test your level of recall or physical skill on your own. Try to find where your ability drops off It takes only a few minutes and is immensely valuable.

You'll learn faster and retain more of what you learn.